# EPS Series

18-slot PXI Express Chassis

**Error Code & Diagnostics**

| | |
|---|---|
| **Manual Rev.:** | 2.00 |
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**Advance Technologies; Automate the World.**

# Revision History

| Revision | Release Date | Description of Change(s) |
|---|---|---|
| 2.00 | 2015/9/30 | Initial Release |

# Preface

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## Conventions

Take note of the following conventions used throughout this manual to make sure that users perform certain tasks and instructions properly.

| | |
|---|---|
| NOTE: | Additional information, aids, and tips that help users perform tasks. |
| CAUTION: | Information to prevent ***minor*** physical injury, component damage, data loss, and/or program corruption when trying to complete a task. |
| WARNING: | Information to prevent ***serious*** physical injury, component damage, data loss, and/or program corruption when trying to complete a specific task. |

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# 1 Introduction

The ADLINK complete EtherCAT solution, from hardware to middleware to software, with every element tailored for dedicated EtherCAT functionality, includes Talos master controllers, EPS slave systems, and remote monitoring and control providers. As well, ADLINK's Softmotion one-stop control kernel delivers flexible and easy-to-use intelligent platforms for driving next-generation modern Smart Factories.

ADLINK's EPS slave system brings the power the flexibility to EtherCAT environments. State-of-the-art modular construction allows the EtherCAT slave to be configured to fulfill the needs of both small local applications and expansive distributed platforms. The EPS system monitors and controls nearly any type of I/O device and sensor as well as motors, with comprehensive and high-density I/O modules that include DI/O, AI/O, and thermal as well as pulse-chain motion modules. Leveraging ARM and FPGA technologies, the EPS slave is able to monitor and report system integrity via EtherCAT communication, while relay module lifecycle and signal integrity of analog output are monitored cyclically, and hot swappability reduces downtime significantly.

Uniquely designed for full operability in conditions from -20°C to 60°C,the EPS slave system is also fully compliant with the EN 60068-2 standard for shock and vibration and EN 61000-6 for heavy industrial EMC protection, as well as emission certification.

Corresponding ESI files for each EPS system can be generated thru ADLINK's Windows-based EtherCAT slave utility, LinkMaster-Pro™.

## 1.1 Features

- Compatible with certified 3rd party EtherCAT masters
- Comprehensive slave modules include high-density digital I/O, analog I/O, thermal measurement, and motion control
- EtherCAT COE, FOE protocol support
- Compliance with industrial IEC standard
- Automatic Detection System by LinkMasterPro™ application detects and generates corresponding ESI files automatically
- Intelligent status monitoring for EPS modules
- Hot-swappable to reduce MTTR
- Operable in environments from -20°C to 60°C

NOTE:

The EPS slave system is fully capable of cooperation with not only ADLINK's Talos master controller but also 3rd party EtherCAT master controllers, by importing corresponding ESI file for eaqch individual module.

## 1.2 Specifications

### 1.2.1 EPS-9905 Chassis

| | | |
|---|---|---|
| **System Level** | | |
| Processor | Xillinx Zynq SoC | |
| Memory | RAM | 512 MB DDR3 |
| | Retain Memory #1 | 16MB Flash on Bus Coupler |
| | Retain Memory #2 | 128KB FRAM on every EPS module |
| **Connectivity** | | |
| EtherCAT | 2 EtherCAT ports | |
| Motion Protocol | Compliant with CiA402 | |
| **Miscellaneous Function** | | |
| Monitored Parameters | Relay lifecycle<br>Analog output signal integrity<br>System power integrity<br>Module Status | |
| **Power Supply** | | |
| DC Input | Built-in 9-30 VDC wide-range DC input<br>3P pluggable connectors with latch (GND, V-, V+) | |
| AC Input | Optional 40 W external AC-DC adapter for AC input | |
| **Mechanical** | | |
| Dimensions | 130 (W) x 110 (D) x 105 (H) mm | |
| Weight | 950 g (net weight of EPS-9905) | |
| Mounting | DIN rail | |
| **Environmental** | | |
| Operating Temperature | -20°C to 60°C | |
| Storage Temperature | -40°C to 85°C (excl. HDD/SDD/CFAST) | |
| Humidity | Approx. °C (non-condensing) | |

| | |
|---|---|
| Vibration | EN 60068-2-6 certification<br>Operating, 5 Grms, 5-500 Hz, 3 axes |
| ESD | Contact +/-4 kV and Air +/-8 kV |
| Shock | EN 60068-2-27 certification<br>Operating, 100 G, half sine 11 ms duration (w/ mSATA) |
| EMC | EN 61000-2 / EN 61000-4 certification<br>CE and FCC Class A |
| Safety | UL, CB |

### 1.2.2 EPS-1132 Module

| | |
|---|---|
| Channels | 32 |
| Input Type | Sinking |
| EVCC Input | 24V ±10% |
| OFF State Input Voltage | <5V (IEC61131-2 Type1/3) |
| OFF State Input Current | <2.3mA |
| ON State Input Voltage | 10V to 30V (IEC61131-2 Type1/3) |
| ON State Input Current | 2.3mA (typ.) |
| IO Protection Input Voltage | Up to 30V |
| IO Protection Reverse Voltage | Up to -30V |
| IO Protection Current limit | < 2.3mA (typ.) |
| Response Time | 150µs to 3ms |
| Power Consumption from Chassis | < 600mW (max.) |
| Thermal Dissipation (at max temp.) | < 2.55W (max.) @ EVCC=24V+10% |
| Isolation (Channel-to-DGND) | 2kV (DC) |

### 1.2.3 EPS-2032 Module

| | |
|---|---|
| Channels | 32 |
| Output Type | Sourcing |
| EVCC Input | 12 to 24V ±10% |
| Output Impedance (Rout) | 200Ω |

| | |
|---|---|
| Output Current (Iout) | 300mA per CH<br>800mA per 4-CH |
| Output Voltage (Vout) | EVCC – (Iout x Rout) |
| IO Protection Reverse Voltage | None |
| IO Protection Current limit | 1.2A for 4CH |
| IO Protection Short Circuit | None |
| Power On State | Off |
| Response Time | 4kHz |
| Power Consumption: Chassis | < 400mW (max.) |
| Thermal Dissipation (at max temp.) | < 2.55W (max.) @ EVCC=24V+10% |
| Isolation (Channel-to-DGND) | 2kV (DC) |

### 1.2.4 EPS-2132 Module

| | |
|---|---|
| Channels | 32 |
| Output Type | Sinking |
| EVCC Input | 12 to 24V ±10% |
| Output Impedance (Rout) | 250Ω |
| Output Current (Iout) | 300mA per CH<br>800mA per 4-CH |
| Output Voltage (Vout) | Iout x Rout |
| IO Protection Reverse Voltage | None |
| IO Protection Current limit | 1.2A for 4CH |
| IO Protection Short Circuit | None |
| Power On State | Off |
| Response Time | 4kHz |
| Power Consumption from Chassis | < 450mW (max.) |
| Thermal Dissipation (at max temp.) | < 2.55W (max.) @<br>EVCC=24V+10% |
| Isolation (Channel-to-DGND) | 2kV (DC) |
| Isolation (Channel-to-Chassis) | 2kV (DC) |

### 1.2.5 EPS-2308 Module

| | |
|---|---|
| Channels | 8 |
| Type | SPST (Single pole single throw) |
| Maximum switched voltage and current | 30V/2A, 240V/0.5A |
| Output Impedance (Rout) | 50mΩ (Typ.) |
| IO Protection Reverse Voltage | Yes |
| Power On State | OFF |
| Response Time | Operate 10ms & release 5ms (Typ.) |
| Power Consumption from Chassis | < 600mW (max.) |
| Thermal Dissipation (at max temp.) | < 2.55W (max.) @ EVCC=24V+10% |
| Isolation (Channel-to-DGND) | DC 2kV 1min. |
| Isolation (Channel-to-Chassis) | DC 2kV 1min. |
| Isolation (Channel-to-Channel) adjacent relay | 250Vrms 1min. |
| Isolation (Channel-to-Channel) adjacent pin within a single relay | 250Vrms 1min. |

### 1.2.6 EPS-3032 Module

| | |
|---|---|
| Channels | 32 Single-ended<br>16 Differential |
| Input Range | ±10V |
| Resolution | 16 bit |
| Offset Error | ±1mV |
| Offset Drift | 0.0004% of range per °C |
| Gain Error | ±0.05% of FSR |
| Gain Drift | 0.002%FSR per °C |
| Voltage Protection | Power On @ ±24V |
| Sampling Rate | 100kHz |
| Power Consumption from Chassis | < 450mW (max.) |

| | |
|---|---|
| Thermal Dissipation (at max temp.) | < 2.55W (max.) @ EVCC=24V+10% |
| Operating Temperature | -20 to 60°C |
| Isolation (Channel-to-DGND) | 2kV (DC) |
| Isolation (Channel-to-Chassis) | 2kV (DC) |

### 1.2.7 EPS-3216 Module

| | |
|---|---|
| Channels | 16 Single-ended |
| Input Range | 0-20mA |
| Resolution | 16 bit |
| Offset Error | 0.5μA |
| Offset Drift | 0.0005% of range per °C |
| Gain Error | ±0.05% of FSR |
| Gain Drift | 0.0008%FSR per °C |
| Voltage Protection | Power On @ ±24V |
| Sampling Rate | 100kHz |
| Power Consumption from Chassis | < 450mW (max.) |
| Thermal Dissipation (at max temp.) | < 2.55W (max.) @ EVCC=24V+10% |
| Operating Temperature | -20 to 60°C |
| Isolation (Channel-to-DGND) | 2kV (DC) |
| Isolation (Channel-to-Chassis) | 2kV (DC) |

### 1.2.8 EPS-3504 Module

| | |
|---|---|
| Channels | 4 RTD |
| Sensor Type | PT100, PT500, PT1000 |
| Technology | 2/3/4-Wire |
| Resolution | 24 bit |
| Sampling Rate | 5 to 20Hz |
| Temperature Range | -200°C to +850°C |
| Path Resistance | 18 to 3900Ω |

| | |
|---|---|
| Accuracy (DC) | ± 0.007% of FSR |
| Accuracy (Temperature) | ± 0.5°C |
| Excitation Current (Max.) | 500μA |
| Power Consumption from Chassis | < 450mW (max.) |
| Thermal Dissipation (at max temp.) | < 2.55W (max.) @ EVCC=24V+10% |
| Operating Temperature | -20 to 60°C |
| Isolation (Channel-to-DGND) | 2kV (DC) |
| Isolation (Channel-to-Chassis) | 2kV (DC) |

### 1.2.9 EPS-4008 Module

| | |
|---|---|
| Channels | 8 Single-ended |
| Output Range | ±10V |
| Resolution | 16 bit |
| Sampling Rate | 100kHz |
| Offset Error | ± 0.2mV |
| Gain Error | ± 0.05% of FSR |
| Offset Drift | ± 0.0075mV /°C |
| Gain Drift | ± 0.00025% /°C of FSR |
| Output Current Capacity | 5 mA |
| Power On State | Relay off |
| Voltage Protection | ± 24V / Relay off when error voltage occurs |
| Power Consumption from Chassis | < 450mW (max.) |
| Thermal Dissipation (at max temp.) | < 2.55W (max.) @ EVCC=24V+10% |
| Operating Temperature | -20 to 60°C |
| Isolation (Channel-to-DGND) | 2kV (DC) |
| Isolation (Channel-to-Chassis) | 2kV (DC) |

### 1.2.10 EPS-6000 Module

| | |
|---|---|
| EPS Module Support (slots) | 4 (Max.) |

| | |
|---|---|
| Field Bus Connectivity | EtherCAT |
| Data Transmission Rate | 100 Mbaud |
| Bus Interface | 2 x RJ45 |
| Cable for EtherCAT Connectivity | CAT5 / CAT5e (recommended) |
| Inner Bus Synchronization | > 0.1μs |
| Supply Voltage | 24VDC (± 10%) |
| Thermal Dissipation (at max temp.) | < 6.6W (max.) @ EVCC=24V+10% |
| Operating Temperature | -20 to 60°C |
| Isolation (Channel-to-DGND) | 2kV (DC) |
| Isolation (Channel-to-Chassis) | 2kV (DC) |

### 1.2.11 EPS-7002 Module

| | |
|---|---|
| Channels | 2 |
| Output Frequency | 4MHz |
| Output Modes | CW/CCW, OUT/DIR |
| Encoder Input Channels | 2 |
| Encoder Input Frequency | 20MHz @ 4xAB |
| Encoder Input Modes | CW/CCW, 1/2/4x AB Phase |
| Encoder Input Type | TTL, Incremental |
| Motion I/O Interface | PEL, MEL, ORG |
| Servo I/O Interface | ALM, INP, RDY, SVON, ERC, RST |
| Input Type | Sinking |
| Input Current | IEC 61131-2, Type 1/3 |
| Output Current (Single Node) (Sinking / Sourcing) | 50mA |
| Power Consumption from Chassis | < 450mW (max.) |
| Thermal Dissipation (at max temp.) | < 2.55W (max.) @ EVCC=24V+10% |
| Operating Temperature | -20 to 60°C |
| Isolation (Channel-to-DGND) | 2kV (DC) |
| Isolation (Channel-to-Chassis) | 2kV (DC) |

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# 2 Error Code & Diagnostics

## 2.1 Error Code and LED Status

EtherCAT device error codes are shown as follows, in which higher priority error codes override lower priority.

| Definition | EPS Module | Pri. | Index 0x1001 | Index 0x1003 | LED | Solution |
|---|---|---|---|---|---|---|
| EtherCAT ID mismatch (among vendor ID/product code/revision number/serial number) | 6000 | 1 | Bit [7] | 0xFF06 | ERR lit | Use LinkMaster Pro to reconfigure settings |
| EtherCAT link is broken (port 0) | 6000 | 2 | Bit [4] | 0x8101 | RJ45 unlit | Check EtherCAT port of EPS-6000. |
| Connection error for internal modules | 6000 | 3 | Bit [7] | 0xFF01 | ERR lit | Remove and reinsert the modules |
| Connection voltage error too high | 6000 | 4 | Bit [2] | 0x3212 | ERR lit | Check setting (index 0x1004, sub-index 1 to 3) and voltage. |
| Connection voltage error too low | 6000 | 5 | Bit [2] | 0x3222 | ERR lit | Check setting (index 0x1004, sub-index 1 to 3) and voltage. |

| Definition | EPS Module | Pri. | Index 0x1001 | Index 0x1003 | LED | Solution |
|---|---|---|---|---|---|---|
| Main device junction voltage error too high | 6000 | 6 | Bit [2] | 0x3211 | ERR lit | Check setting (index 0x1004, sub-index 4 to 6) and voltage. |
| Main device junction voltage error too low | 6000 | 7 | Bit [2] | 0x3221 | ERR lit | Check setting (index 0x1004, sub-index 4 to 6) and voltage. |
| BC connector current error too high | 6000 | 8 | Bit [1] | 0x2211 | ERR lit | Check setting (index 0x1004, sub-index 7 to 9) and current. |
| BC connector current error too low (between EPS-6000 and EPS-9805) | 6000 | 9 | Bit [1] | 0x2211 | ERR lit | Check setting (index 0x1004, sub-index 7 to 9) and current. |
| Module current error (between EPS-9805 and all modules exceeds 1A) | 6000 | 10 | Bit [1] | 0x2222 | POWER unlit (All) | No action necessary (restored when temperature falls) |
| Internal handshake error for modules (Communication times out (100 μs) between EPS-6000 and other modules.) | 6000 | 11 | Bit [7] | 0xFF02 | ERR lit | Check connection between EPS-6000 and other modules. |

| Definition | EPS Module | Pri. | Index 0x1001 | Index 0x1003 | LED | Solution |
|---|---|---|---|---|---|---|
| EPS-7002 emergency | 7002 | 12 | Bit[7] | 0xFF03 | ERR lit | Solve servo drive problem. |
| EPS-7002 alarm | 7002 | 13 | Bit[7] | 0xFF04 | ERR lit | Solve servo drive problem. |
| EPS-7002 no communication error (5 second timeout with EPS-6000) | 7002 | 14 | Bit[4] | 0x8102 | ERR lit | Solve EPS-7002 problem or check connection between EPS-6000 and EPS-7002 (index 0x1004, sub-index 10). |
| EPS-2032/EPS-2132 no communication error (5 second timeout with EPS-6000) | 2032/ 2132 | 15 | Bit[4] | 0x8103 | ERR lit (both) | Solve EPS-2032/2132 problem or check connection between EPS-6000 and EPS-2032/EPS-2132 (index 0x1004, sub-index 11). |
| EPS-2308 no communication error (5 second timeout with EPS-6000) | 2308 | 16 | Bit[4] | 0x8104 | ERR lit | Solve EPS-2308 problem or check connection between EPS-6000 and EPS-2308(index 0x1004, sub-index 12). |

| Definition | EPS Module | Pri. | Index 0x1001 | Index 0x1003 | LED | Solution |
|---|---|---|---|---|---|---|
| EPS-4008 no communication error (5 second timeout with EPS-6000) | 4008 | 17 | Bit[4] | 0x8105 | ERR lit | Solve theEPS-4008 problem or check connection between EPS-6000 and EPS-4008(index 0x1004, sub-index 13). |
| Module temperature too high | 6000 | 18 | Bit[3] | 0x4212 | ERR lit (All) | Check settings (index 0x1004, sub-index 14 to 16) and temperature of each module. |
| Module temperature too low | 6000 | 19 | Bit[3] | 0x4222 | ERR lit (All) | Check settings (index 0x1004, sub-index 14 to 16) and temperature of each module. |
| Main device junction temperature warning too high | 6000 | 20 | Bit[3] | 0x4211 | ERR lit | Check settings (index 0x1004, sub-index 17 to 19) and temperature. |

| Definition | EPS Module | Pri. | Index 0x1001 | Index 0x1003 | LED | Solution |
|---|---|---|---|---|---|---|
| Main device junction temperature warning too low | 6000 | 21 | Bit[3] | 0x4221 | ERR lit (All) | Check settings (index 0x1004, sub-index 17 to 19) and temperature. |
| EPS-2308 counter expired warning | 2308 | 22 | Bit[7] | 0xFF05 | ERR blinking | Check the setting (index 0x1004, sub-index 20) to enable/ disable the checker. |
| EPS-1132 over temperature warning (over 150°) | 1132 | 23 | Bit[3] | 0x4213 | ERR blinking | Check the setting (index 0x1004, sub-index 21) to enable/ disable the checker. |
| EPS-4008 voltage difference error too high | 4008 | 24 | Bit[2] | 0x3111 | ERR lit | Check the setting (index 0x1004, sub-index 22 to 24) and voltage difference of EPS-4008. |

| Definition | EPS Module | Pri. | Index 0x1001 | Index 0x1003 | LED | Solution |
|---|---|---|---|---|---|---|
| EPS-4008 voltage difference error too low | 4008 | 25 | Bit[2] | 0x3121 | ERR lit | Check the setting (index 0x1004, sub-index 22 to 24) and voltage difference of EPS-4008. |
| EPS-6000 SyncManager Output Sync Error (index 0x1C32:32 is 1, indicating synch error) | 6000 | 27 | Bit[4] | 0x8106 | ERR lit | Check the setting (index 0x1004, sub-index 25) and cycle time. |
| EPS-6000 SyncManager Input Sync Error (index 0x1C33:32 is 1, indicating synch error) | 6000 | 28 | Bit[4] | 0x8107 | ERR lit | Check the setting (index 0x1004, sub-index 26) and cycle time. |
| Parameter Error (input parameter illegal) | 6000 | 29 | Bit[7] | 0xFF80 | ERR lit | Check parameter setting during operation. |

## 2.2 EPS Module LED Status

Each module's LED display (listed left to right) reports status as follows.

NOTE: EPS-2308 (RO), provides no status LEDs.

| Module | LED | Display | Status |
|---|---|---|---|
| EPS-6000 (BC) | PWR | Lit | Power on |
| | | Unlit | Power off |
| | ERR | Lit | Error, check error code |
| | | Unlit | Normal |
| | | Blinking | Warning, check error code |
| | RUN | Lit | Normal |
| | | Unlit | Watchdog timeout |
| EPS-1132 (DI32) | STS | Lit | Error, check error code |
| | | Unlit | Normal |
| | | Blinking | Warning, check error code |
| | PWR | Lit | Power on |
| | | Unlit | Power off |
| EPS-2032 (DO32) | STS | Lit | Error, check error code |
| | | Unlit | Normal |
| | PWR | Lit | Power on |
| | | Unlit | Power off |
| EPS-2132 (DO32) | STS | Lit | Error, check error code |
| | | Unlit | Normal |
| | PWR | Lit | Power on |
| | | Unlit | Power off |
| EPS-3032 (AI32) | STS | Lit | Error, check error code |
| | | Unlit | Normal |
| | PWR | Lit | Power on |
| | | Unlit | Power off |
| EPS-3216 (AI16) | STS | Lit | Error, check error code |
| | | Unlit | Normal |
| | PWR | Lit | Power on |
| | | Unlit | Power off |

| Module | LED | Display | Status |
|---|---|---|---|
| EPS-3504 (RTD) | STS | Lit | Error, check error code |
| | | Unlit | Normal |
| | PWR | Lit | Power on |
| | | Unlit | Power off |
| EPS-4008 (AO8) | STS | Lit | Error, check error code |
| | | Unlit | Normal |
| | PWR | Lit | Power on |
| | | Unlit | Power off |
| EPS-7002 (MO) | EMG | Lit | Error, external emergency |
| | | Unlit | Normal |
| | STS | Lit | Error, check error code |
| | | Unlit | Normal |
| | PWR | Lit | Power on |
| | | Unlit | Power off |

## 2.3 Error History & Log

Index 0x1001 and 0x1003 error histories are defined in CIA301, in sequence as follows.

1. If LEDs or index 0x1001 do not equal 0x0000, an error or warning occurs.
2. In index 0x1003, the highest priority error code is represented in sub-index 0x01, with history available in other sub-indices.
3. When the EPS system problem is solved, check for other lower priority errors.

For EPS-7002, index 0x603F lists related errors.

The index 0x1005 Error Log can be seen via the Error Log tab for EPS-6000 in LinkMasterPro, or viewed as follows.

- ▶ Set 0x1005:01 to enable the logger. For the logs, total depth = 256 can be stored in non-volatile memory. Address 0 indi-

cates the latest error log. If more than 256 error logs list, the oldest will be overwritten. Setting 0x1005:02 clears all logs.

- 0x1005:03 lists the valid count of logs.
- Setting 0x1005:04 shows error content of specified addresses.

Reading 0x1005:05 and 0x1005:06 allows construction of a 64-bit error log. 0x1005 provides definitions of each bit.

## 2.4 Object Dictionary Summary

### 2.4.1 EPS-6000 Object Dictionary Section Summary

| Indexes | Section Name | Description |
|---|---|---|
| 0x1000 to 0x1FFF | Communication Objects | Defined in CIA301. |
| 0x2000 to 0x2FFF | Manufacturer-Specific Area 1 | Allocates size 0x200 for each axis which follows CIA402. |
| 0x3000 to 0x3FFF | Input Area | Allocates size 0x400 for each module which follows ETG. 5001. The ODs store TxPDO. |
| 0x4000 to 0x4FFF | Output Area | Allocates size 0x400 for each module which follows ETG. 5001. The ODs are used to store RxPDO. |
| 0x5000 to 0x5FFF | Reserved | N/A |
| 0x6000 to 0x9FFF | CIA402 Area | Allocates size 0x800 for each axis which follows CIA402. |
| 0xA000 to 0xBFFF | Reserved | N/A |
| 0xC000 to 0xCFFF | Manufacturer-Specific Area 2 | Allocates size 0x400 for each module which follows ETG. 5001. The ODs store manufacturer-specific parameters. |
| 0xDFFF to 0xEFFF | Reserved | N/A |
| 0xF000 to 0xFFFF | Device Area | Follows ETG. 5001. The ODs store MDP device parameters. |

### 2.4.2 EPS-6000

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x1000 | VAR | Device Type | UNSIGNED32 | RO |
| 0x1001 | VAR | Error Register | UNSIGNED8 | RO |
| 0x1003 | ARRAY | Error History | UNSIGNED8 | RW |
| 0x1004 | ARRAY | Error Range Setting | UNSIGNED8 | RW |
| 0x1005 | ARRAY | Error Log | UNSIGNED8 | RW |
| 0x1008 | VAR | Device Name | STRING | RO |
| 0x1009 | VAR | Hardware Version | STRING | RO |
| 0x100A | VAR | Software Version | STRING | RO |
| 0x100B | VAR | Manufactory Version | UNSIGNED8 | RO |
| 0x100C | ARRAY | Scan Module | UNSIGNED8 | RW |
| 0x100F | VAR | FOE Status | UNSIGNED16 | RO |
| 0x1018 | ARRAY | Identity | UNSIGNED8 | RO |
| 0x10F1 | ARRAY | Error Handling | UNSIGNED8 | RW |
| 0x10F3 | ARRAY | Diagnosis History | UNSIGNED8 | RW |
| 0x1100 | ARRAY | PDO Numbers | UNSIGNED8 | RO |
| 0x1313 | ARRAY | Serial Number Setting | UNSIGNED8 | RW |
| 0x1600 | ARRAY | Slot0_RxPDO-Map_[NNNN] | UNSIGNED8 | RO |
| 0x1601 | ARRAY | Slot1_RxPDO-Map_[NNNN] | UNSIGNED8 | RO |
| 0x1602 | ARRAY | Slot2_RxPDO-Map_[NNNN] | UNSIGNED8 | RO |
| 0x1603 | ARRAY | Slot3_RxPDO-Map_[NNNN] | UNSIGNED8 | RO |
| 0x1604 | ARRAY | SlotB_RxPDO-Map_6000 | UNSIGNED8 | RO |
| 0x1A00 | ARRAY | Slot0_TxPDO-Map_[NNNN] | UNSIGNED8 | RO |
| 0x1A01 | ARRAY | Slot1_TxPDO-Map_[NNNN] | UNSIGNED8 | RO |
| 0x1A02 | ARRAY | Slot2_TxPDO-Map_[NNNN] | UNSIGNED8 | RO |
| 0x1A03 | ARRAY | Slot3_TxPDO-Map_[NNNN] | UNSIGNED8 | RO |
| 0x1A04 | ARRAY | SlotB_TxPDO-Map_6000 | UNSIGNED8 | RO |
| 0x1C00 | ARRAY | Sync manger type | UNSIGNED8 | RO |
| 0x1C12 | ARRAY | RxPDO assign | UNSIGNED8 | RO |
| 0x1C13 | ARRAY | TxPDO assign | UNSIGNED8 | RO |
| 0x1C32 | ARRAY | SM output parameter | UNSIGNED8 | RW |
| 0x1C33 | ARRAY | SM input parameter | UNSIGNED8 | RW |

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x1C34 | ARRAY | DC Parameter Setting | UNSIGNED8 | RW |
| 0x2000 to 0x2FFF | | In this section, please check summary | | |
| 0x3000 to 0x33FF | | Link to slot0 TxPDO | | |
| 0x3400 to 0x37FF | | Link to slot1 TxPDO | | |
| 0x3800 to 0x3BFF | | Link to slot2 TxPDO | | |
| 0x3C00 to 0x3FFE | | Link to slot3 TxPDO | | |
| 0x3FFF | VAR | SlotB_Input_6000 | UNSIGNED32 | RO |
| 0x4000 to 0x43FF | | Link to slot0 RxPDO | | |
| 0x4400 to 0x47FF | | Link to slot1 RxPDO | | |
| 0x4800 to 0x4BFF | | Link to slot2 RxPDO | | |
| 0x4C00 to 0x4FFE | | Link to slot3 RxPDO | | |
| 0x4FFF | VAR | SlotB_Output_6000 | UNSIGNED32 | RW |
| 0x6000 to 0x9FFF | | In this section, please check the summary | | |
| 0xC000 to 0xC3FF | | Link to slot0 | | |

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0xC400 to 0xC7FF | | Link to slot1 | | |
| 0xC800 to 0xCBFF | | Link to slot2 | | |
| 0xCC00 to 0xCFFF | | Link to slot3 | | |
| 0xF000 | ARRAY | Modular Device Profile | UNSIGNED8 | RO |
| 0xF010 | ARRAY | Module Profile List | UNSIGNED8 | RO |
| 0xF030 | ARRAY | Configured Module Ident List | UNSIGNED8 | RO |
| 0xF050 | ARRAY | Detected Module Ident List | UNSIGNED8 | RO |

### 2.4.3 EPS-1132

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x3000 | VAR | Slot[N]_DI-Data_1132 | UNSIGNED32 | RO |
| 0xC000 | ARRAY | Slot[N]_DI-Info_1132 | UNSIGNED8 | RO |
| 0xC001 | ARRAY | Slot[N]_DI-Config_1132 | UNSIGNED8 | RW |

### 2.4.4 EPS-2032

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x4000 | VAR | Slot[N]_DO-Data_2032 | UNSIGNED32 | RW |
| 0xC000 | ARRAY | Slot[N]_DO-Info_2032 | UNSIGNED8 | RO |
| 0xC001 | ARRAY | Slot[N]_DO-Config_2032 | UNSIGNED8 | RW |

### 2.4.5 EPS-2132

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x4000 | VAR | Slot[N]_DO-Data_2132 | UNSIGNED32 | RW |
| 0xC000 | ARRAY | Slot[N]_DO-Info_2132 | UNSIGNED8 | RO |
| 0xC001 | ARRAY | Slot[N]_DO-Config_2132 | UNSIGNED8 | RW |

### 2.4.6 EPS-3032

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x3000 | ARRAY | Slot[N]_AI-Data_3032 | UNSIGNED8 | RO |
| 0xC000 | ARRAY | Slot[N]_AI-Info_3032 | UNSIGNED8 | RO |
| 0xC001 | ARRAY | Slot[N]_AI-Config_3032 | UNSIGNED8 | RW |
| 0xC002 | ARRAY | Slot[N]_AI-Calibration_3032 | UNSIGNED8 | RW |

### 2.4.7 EPS-3216

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x3000 | ARRAY | Slot[N]_AI-Data_3216 | UNSIGNED8 | RO |
| 0xC000 | ARRAY | Slot[N]_AI-Info_3216 | UNSIGNED8 | RO |

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0xC001 | ARRAY | Slot[N]_AI-Config_3216 | UNSIGNED8 | RW |
| 0xC002 | ARRAY | Slot[N]_AI-Calibration_3216 | UNSIGNED8 | RW |

### 2.4.8 EPS-4008

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x4000 | ARRAY | Slot[N]_AO-Data_4008 | UNSIGNED8 | RW |
| 0xC000 | ARRAY | Slot[N]_AO-Info_4008 | UNSIGNED8 | RO |
| 0xC001 | ARRAY | Slot[N]_AO-Config_4008 | UNSIGNED8 | RW |
| 0xC002 | ARRAY | Slot[N]_AO-Calibration_4008 | UNSIGNED8 | RW |

### 2.4.9 EPS-3504

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x3000 | ARRAY | Slot[N]_RTD-Data_3504 | UNSIGNED8 | RO |
| 0xC000 | ARRAY | Slot[N]_RTD-Info_3504 | UNSIGNED8 | RO |
| 0xC001 | ARRAY | Slot[N]_RTD-Config_3504 | UNSIGNED8 | RW |

### 2.4.10 EPS-2308

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x4000 | VAR | Slot[N]_RO-Data_2308 | UNSIGNED32 | RW |
| 0xC000 | ARRAY | Slot[N]_RO-Info_2308 | UNSIGNED8 | RO |
| 0xC001 | ARRAY | Slot[N]_RO-Config_2308 | UNSIGNED8 | RW |

### 2.4.11 EPS-7002

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x2010 | VAR | Slot[N]_Reset-Group_7002 | UNSIGNED32 | RW |
| 0x2020 | VAR | Slot[N]_Error-Status_7002 | UNSIGNED32 | RW |
| 0x2024 | VAR | Slot[N]_Error-Enable_7002 | UNSIGNED32 | RW |
| 0x210C | VAR | Slot[N]_Output-Counter_7002 | UNSIGNED32 | RW |
| 0x2110 | VAR | Slot[N]_MO-IO-Filter-Enable_7002 | UNSIGNED32 | RW |
| 0x2114 | VAR | Slot[N]_MO-IO-Filter-Polarity_7002 | UNSIGNED32 | RW |
| 0x2118 | VAR | Slot[N]_Encoder-Config_7002 | UNSIGNED32 | RW |
| 0x230C | VAR | Slot[N]_DDA-Cycle-Time_7002 | UNSIGNED32 | RW |
| 0x2310 | VAR | Slot[N]_Axis-0-Current-Position_7002 | INTEGER32 | RO |
| 0x2314 | VAR | Slot[N]_Axis-1-Current-Position_7002 | INTEGER32 | RO |
| 0x2318 | VAR | Slot[N]_Axis-0-Ez-Latch_7002 | UNSIGNED32 | RW |
| 0x231C | VAR | Slot[N]_Axis-1-Ez-Latch_7002 | UNSIGNED32 | RW |
| 0x2320 | VAR | Slot[N]_Axis-0-ORG-Latch_7002 | UNSIGNED32 | RW |
| 0x2324 | VAR | Slot[N]_Axis-1-ORG-Latch_7002 | UNSIGNED32 | RW |
| 0x2328 | VAR | Slot[N]_Axis-0-PEL-Latch_7002 | UNSIGNED32 | RW |
| 0x232C | VAR | Slot[N]_Axis-1-PEL-Latch_7002 | UNSIGNED32 | RW |
| 0x2330 | VAR | Slot[N]_Axis-0-MEL-Latch_7002 | UNSIGNED32 | RW |
| 0x2334 | VAR | Slot[N]_Axis-1-MEL-Latch_7002 | UNSIGNED32 | RW |
| 0x2340 | VAR | Slot[N]_Motion-Input-Noise_7002 | UNSIGNED32 | RW |
| 0x2354 | VAR | Slot[N]_Trigger-Ctrl_7002 | UNSIGNED32 | RW |
| 0x2378 | VAR | Slot[N]_Axis-0-Internal-Current-Position_7002 | INTEGER32 | RO |
| 0x237C | VAR | Slot[N]_Axis-1-Internal-Current-Position_7002 | INTEGER32 | RO |
| 0x603F | VAR | Slot[N]_Axis-0-Error-Code_7002 | UNSIGNED16 | RO |
| 0x6040 | VAR | Slot[N]_Axis-0-Control-Word_7002 | UNSIGNED32 | RW |
| 0x6041 | VAR | Slot[N]_Axis-0-Status-Word_7002 | UNSIGNED32 | RO |
| 0x6060 | VAR | Slot[N]_Axis-0-Modes-Of-Operation_7002 | INTEGER8 | RW |

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x6061 | VAR | Slot[N]_Axis-0-Modes-Of-Operation-Dsplay_7002 | INTEGER8 | RO |
| 0x6064 | VAR | Slot[N]_Axis-0-Position-Actual-Value_7002 | INTEGER32 | RO |
| 0x607A | VAR | Slot[N]_Axis-0-Target-Position_7002 | INTEGER32 | RW |
| 0x60B8 | VAR | Slot[N]_Axis-0-Touch-Probe-Ctrl_7002 | UNSIGNED16 | RW |
| 0x60B9 | VAR | Slot[N]_Axis-0-Touch-Probe-Response_7002 | UNSIGNED32 | RO |
| 0x60BA | VAR | Slot[N]_Axis-0-Touch-Probe-0-Positive-Edge-Value_7002 | UNSIGNED32 | RO |
| 0x60BB | VAR | Slot[N]_Axis-0-Touch-Probe-0-Negative-Edge-Value_7002 | UNSIGNED32 | RO |
| 0x60BC | VAR | Slot[N]_Axis-0-Touch-Probe-1-Positive-Edge-Value_7002 | UNSIGNED32 | RO |
| 0x60BD | VAR | Slot[N]_Axis-0-Touch-Probe-1-Negative-Edge-Value_7002 | UNSIGNED32 | RO |
| 0x60C2 | ARRAY | Slot[N]_Axis-0-Interpolation-Time-Period _7002 | UNSIGNED8 | RW |
| 0x60D0 | VAR | Slot[N]_Axis-0-Touch-Probe-Source_7002 | UNSIGNED16 | RW |
| 0x60FD | VAR | Slot[N]_Axis-0-Motion-Input_7002 | UNSIGNED32 | RO |
| 0x60FE | VAR | Slot[N]_Axis-0-Motion-Output_7002 | UNSIGNED32 | RW |
| 0x6502 | VAR | Slot[N]_Axis-0-Support-Drive-Modes_7002 | UNSIGNED32 | RO |
| 0x683F | VAR | Slot[N]_Axis-1-Error-Code_7002 | UNSIGNED16 | RO |
| 0x6840 | VAR | Slot[N]_Axis-1-Control-Word_7002 | UNSIGNED32 | RW |
| 0x6841 | VAR | Slot[N]_Axis-1-Status-Word_7002 | UNSIGNED32 | RO |
| 0x6860 | VAR | Slot[N]_Axis-1-Modes-Of-Operation_7002 | INTEGER8 | RW |
| 0x6861 | VAR | Slot[N]_Axis-1-Modes-Of-Operation-Dsplay_7002 | INTEGER8 | RO |
| 0x6864 | VAR | Slot[N]_Axis-1-Position-Actual-Value_7002 | INTEGER32 | RO |

| Index | Object | Name | Type | Attr. |
|---|---|---|---|---|
| 0x687A | VAR | Slot[N]_Axis-1-Target-Position_7002 | INTEGER32 | RW |
| 0x68B8 | VAR | Slot[N]_Axis-1-Touch-Probe-Ctrl_7002 | UNSIGNED16 | RW |
| 0x68B9 | VAR | Slot[N]_Axis-1-Touch-Probe-Response_7002 | UNSIGNED32 | RO |
| 0x68BA | VAR | Slot[N]_Axis-1-Touch-Probe-0-Positive-Edge-Value_7002 | UNSIGNED32 | RO |
| 0x68BB | VAR | Slot[N]_Axis-1-Touch-Probe-0-Negative-Edge-Value_7002 | UNSIGNED32 | RO |
| 0x68BC | VAR | Slot[N]_Axis-1-Touch-Probe-1-Positive-Edge-Value_7002 | UNSIGNED32 | RO |
| 0x68BD | VAR | Slot[N]_Axis-1-Touch-Probe-1-Negative-Edge-Value_7002 | UNSIGNED32 | RO |
| 0x68C2 | ARRAY | Slot[N]_Axis-1-Interpolation-Time-Period _7002 | UNSIGNED8 | RW |
| 0x68D0 | VAR | Slot[N]_Axis-1-Touch-Probe-Source_7002 | UNSIGNED16 | RW |
| 0x68FD | VAR | Slot[N]_Axis-1-Motion-Input_7002 | UNSIGNED32 | RO |
| 0x68FE | VAR | Slot[N]_Axis-1-Motion-Output_7002 | UNSIGNED32 | RW |
| 0x6D02 | VAR | Slot[N]_Axis-1-Support-Drive-Modes_7002 | UNSIGNED32 | RO |
| 0xC000 | ARRAY | Slot[N]_MO_Info_7002 | UNSIGNED8 | RO |

# Important Safety Instructions

For user safety, please read and follow all **instructions**, **WARNINGS**, **CAUTIONS,** and **NOTES** marked in this manual and on the associated equipment before handling/operating the equipment.

- Read these safety instructions carefully.
- Keep this user's manual for future reference.
- Read the specifications section of this manual for detailed information on the operating environment of this equipment.
- When installing/mounting or uninstalling/removing equipment:
  - Turn off power and unplug any power cords/cables.
- To avoid electrical shock and/or damage to equipment:
  - Keep equipment away from water or liquid sources;
  - Keep equipment away from high heat or high humidity;
  - Keep equipment properly ventilated (do not block or cover ventilation openings);
  - Make sure to use recommended voltage and power source settings;
  - Always install and operate equipment near an easily accessible electrical socket-outlet;
  - Secure the power cord (do not place any object on/over the power cord);
  - Only install/attach and operate equipment on stable surfaces and/or recommended mountings; and,
  - If the equipment will not be used for long periods of time, turn off and unplug the equipment from its power source.

- Never attempt to fix the equipment. Equipment should only be serviced by qualified personnel.
- A Lithium-type battery may be provided for uninterrupted, backup or emergency power.

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WARNING:

Risk of explosion if battery is replaced with an incorrect type; please dispose of used batteries appropriately.

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- Equipment must be serviced by authorized technicians when:
  - The power cord or plug is damaged;
  - Liquid has penetrated the equipment;
  - It has been exposed to high humidity/moisture;
  - It is not functioning or does not function according to the user's manual;
  - It has been dropped and/or damaged; and/or,
  - It has an obvious sign of breakage.

# Getting Service

Contact us should you require any service or assistance.

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